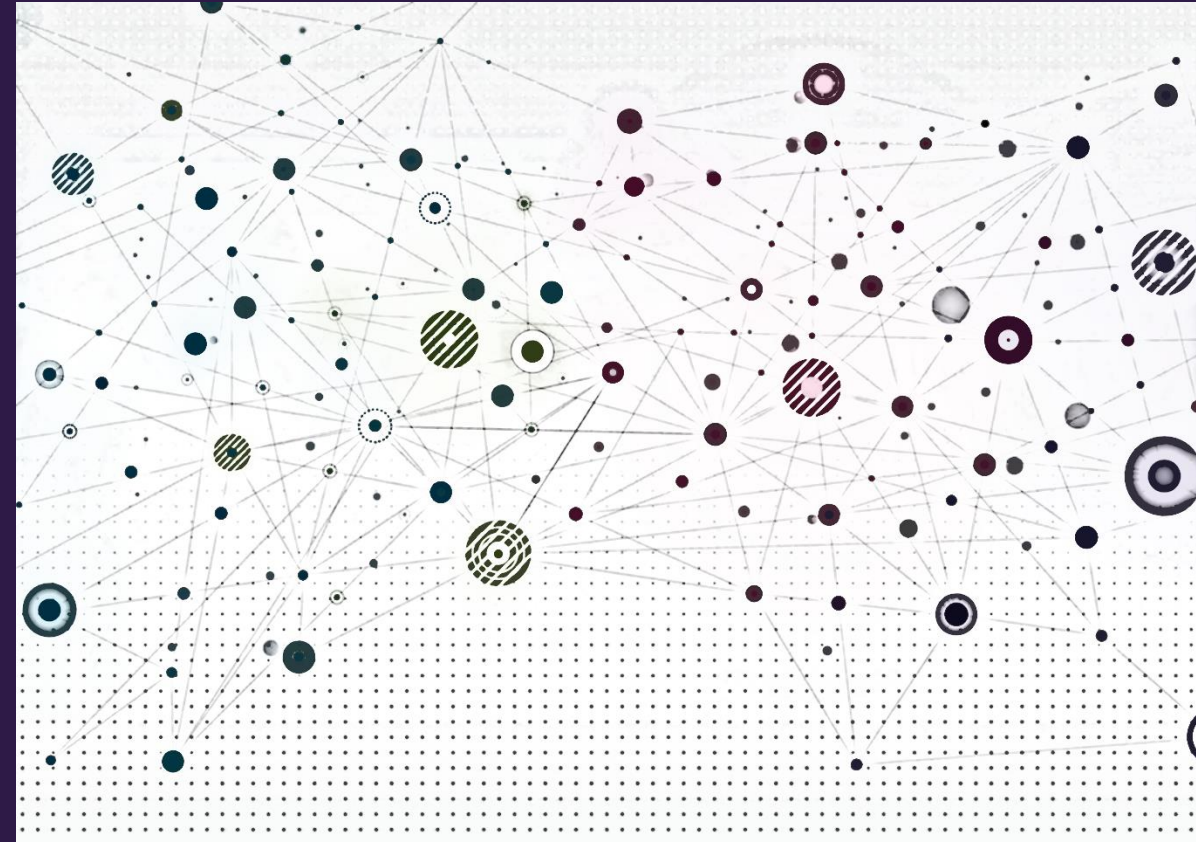


Heterogeneous Graphs





Overview

- Introduction to heterogeneous graphs
- Extending GCNs for heterogeneous graphs
- Inferences on relational graphs



Heterogeneous graph introduction

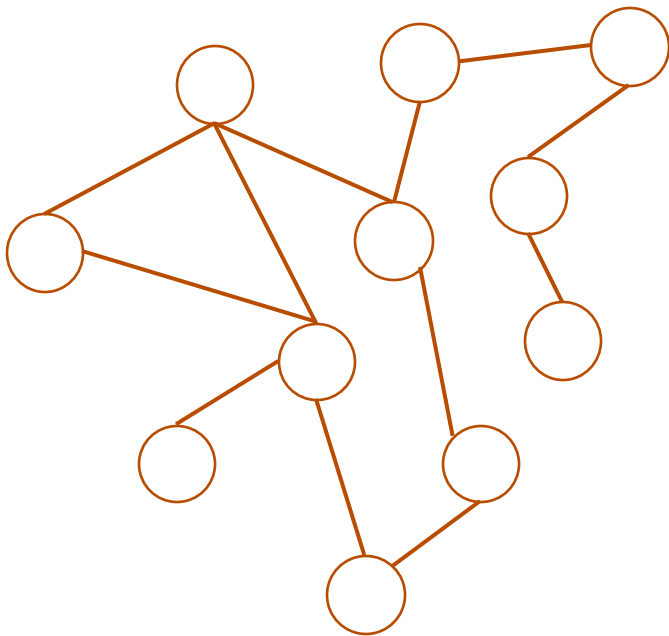




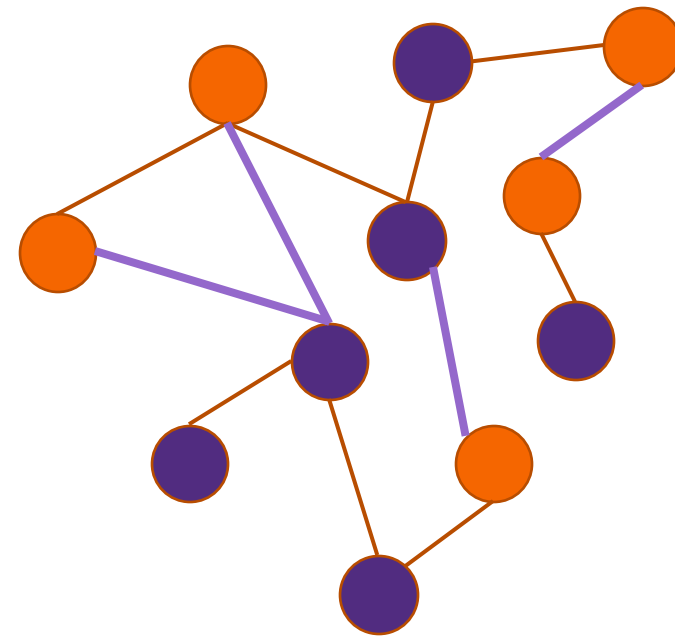
Heterogeneous vs. homogeneous graphs

To date, we have only considered homogeneous graphs

Homogeneous graphs– all nodes and edges are of the same type and have the same attributes



Heterogeneous graph– different node and/or edge types with possibly different attributes.

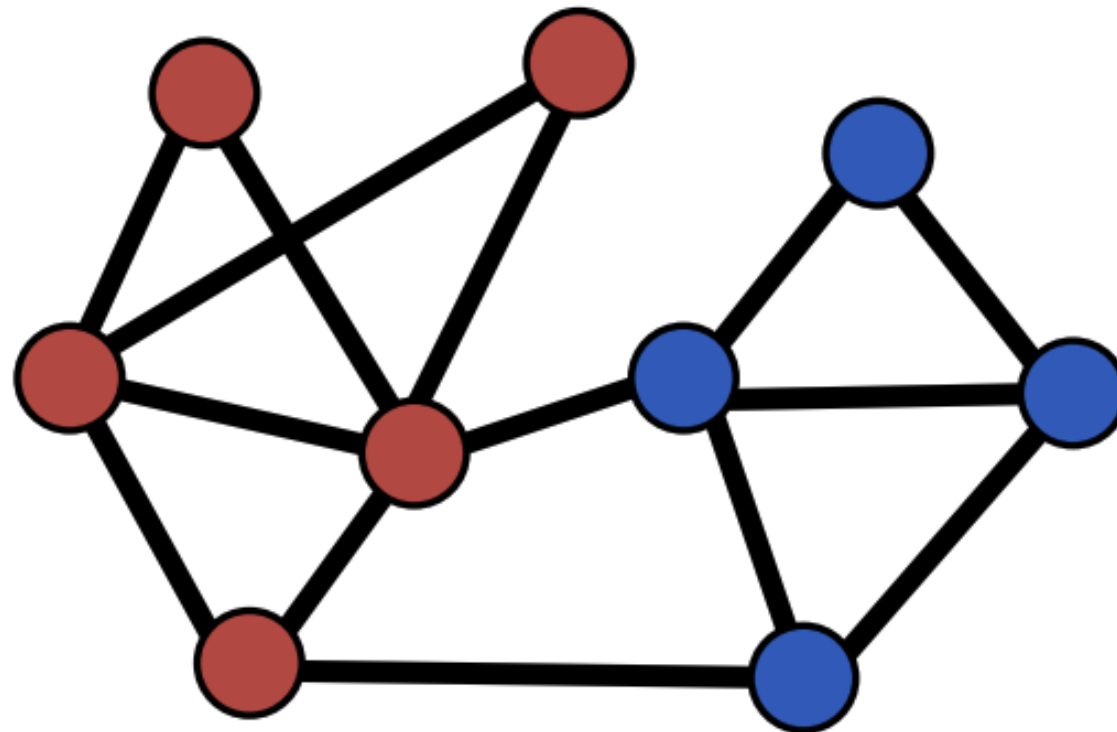




Heterogenous graphs example

Citation network with 2 types of nodes:

- Node **type A**: paper nodes
- Node **type B**: author nodes

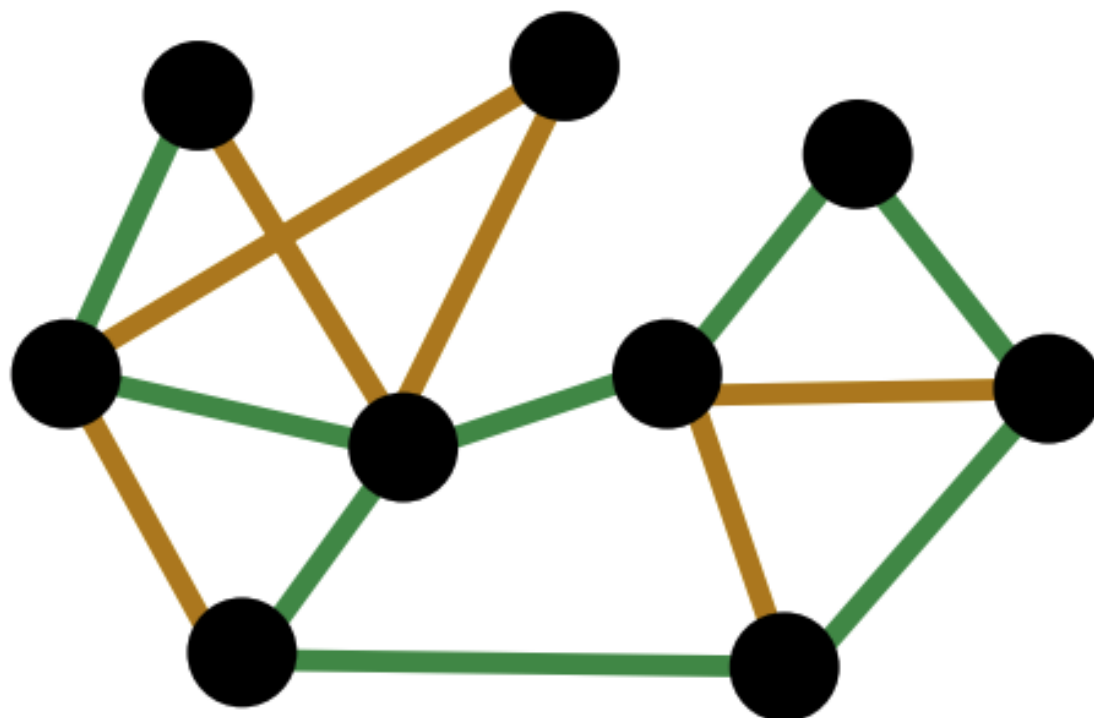




Heterogenous graphs example

Citation network with 2 types of edges:

- Edge **type A**: cites
- Node **type B**: likes





Heterogenous graphs example

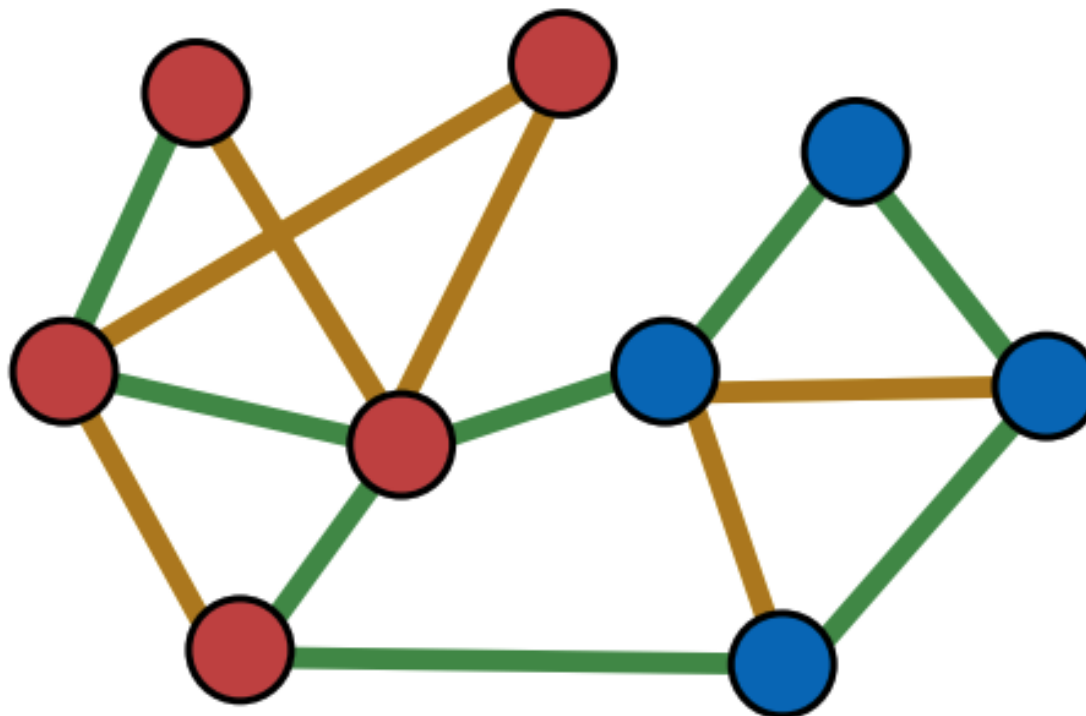
- Overall graph has 2 node and 2 edge types
- 8 possible **relation** types in this graph where the relation type is defined by the triple: (node_start type, edge type, node_end type)

(paper, cites, paper)

(paper, likes, paper)

(paper, cites, author)

(paper, likes, author)



(author, cites, author)

(author, likes, author)

(author, cites, paper)

(author, likes, author)



Heterogeneous graphs terminology

- Define a heterogeneous graph as

$$G = (V, E, \tau, \phi)$$

where

- **Nodes** $v \in V$ have a node type $\tau(v)$
 - Example: $\tau(v) \in \{author, paper\}$
- **Edges** $(u, v) \in E$ have an edge type $\phi(u, v)$
 - Example: $\phi(u, v) \in \{likes, cites\}$
- **Relation** types $r = (\tau(u), \phi(u, v), \tau(v))$
 - Example: (author, cites, paper)
- Different node and edge types can have different feature spaces

Many graphs are heterogeneous

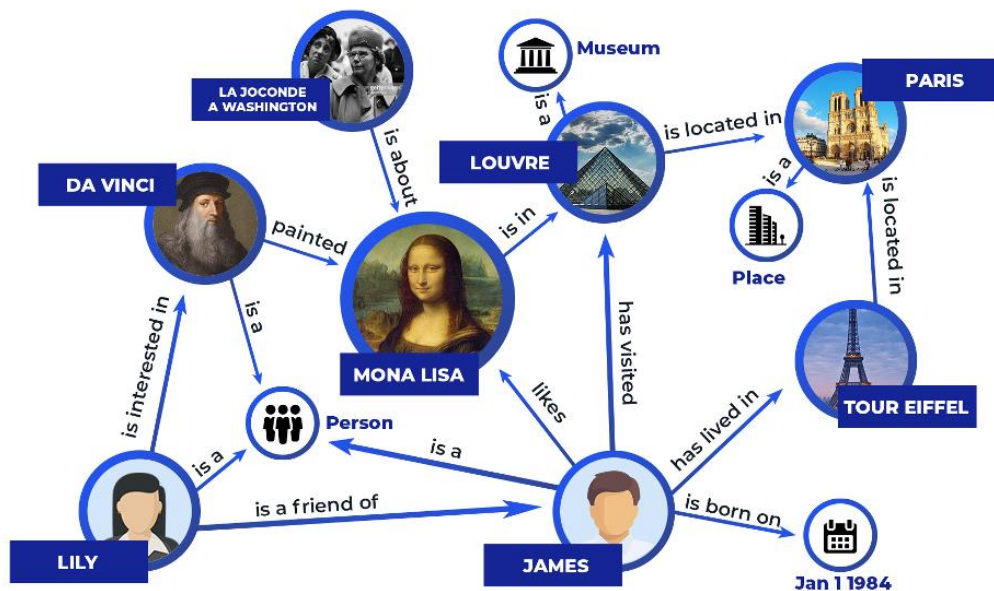


Image credit: Zilliz

Knowledge Graphs

Node types: Artifact, Museum, Person, Place

Edge types: is_a, is_a_friend_of, is_born_on

Example Relation: Lily, is_friend_of, James

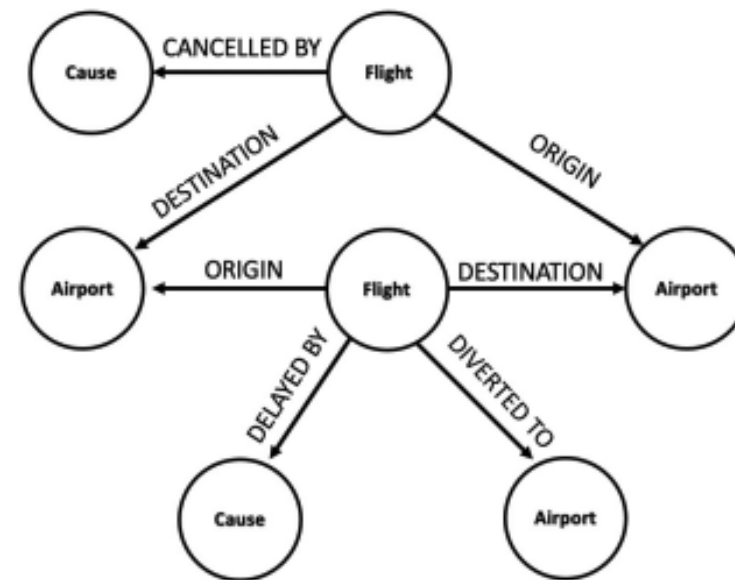


Image credit: Jure Leskovec, <http://cs224w.stanford.edu>

Event Graphs

Node types: Airport, Cause, Flight

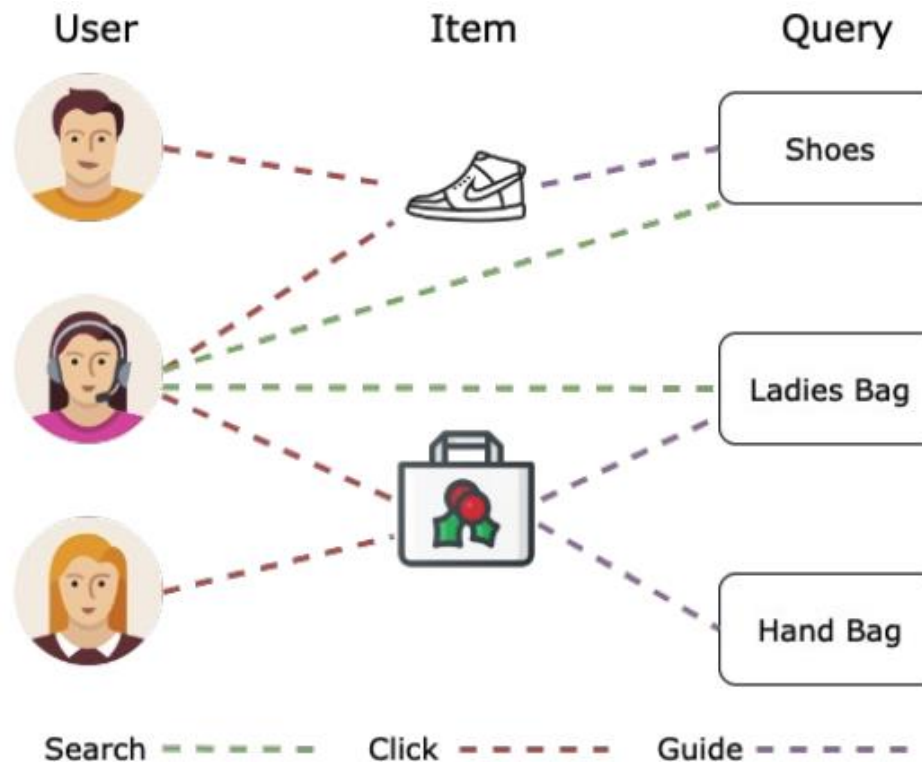
Edge types: Origin, Delayed By, Cancelled By

Example Relation: Flight, Delayed BY, Cause

Many graphs are heterogeneous

E-commerce graph

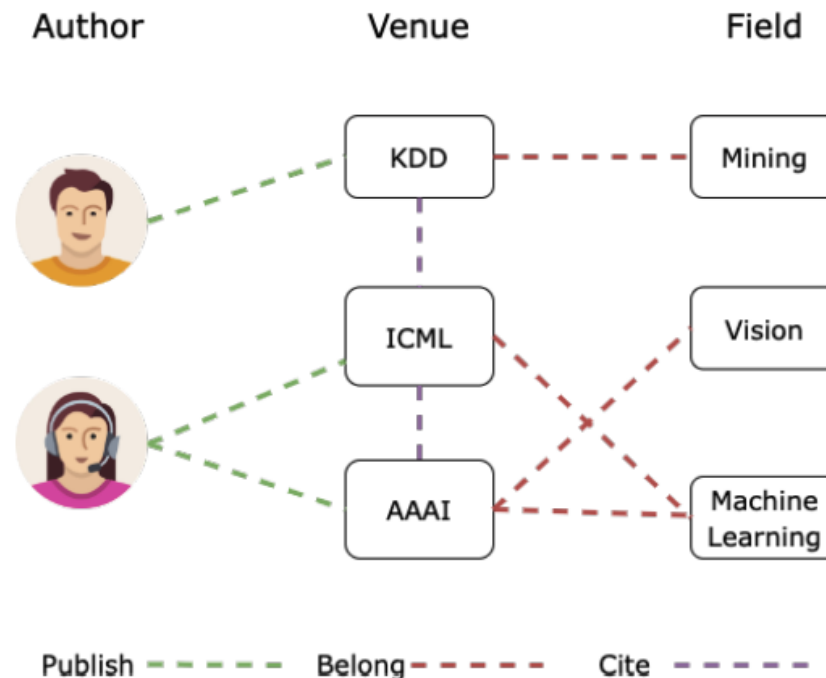
- Node types: User, Item, Query, ...
- Edge types: Purchase, Visit, Guide, Search, ...
- Different node types have different feature spaces



Many graphs are heterogeneous

Microsoft Open Academic Graph

- Node types: Author, Paper, Venue, Field, ...
- Edge types: Publish, Cite, Belong, ...
- >300M papers, >64M edges
- <https://www.microsoft.com/en-us/research/project/open-academic-graph/>





Types as features?

- We can potentially treat node and edge types as features
 - Append a one-hot encoding to the node or edge feature vector
 - Append $[1,0]$ to the feature vector of author nodes and $[0,1]$ for paper nodes
 - Append $[1,0,0]$ to “cites” edges, $[0,1,0]$ to “publish”, and $[0,0,1]$ to “belong”
- Heterogeneous graph reduces to a homogeneous graph in this case
- When do we need a heterogeneous graph?



When do we need a heterogeneous graph?

- Case 1: Different node / edge types have different feature dimensions
 - Example: "Author" nodes have 4 features while "Paper" nodes have 10 features
- Case 2: Different relation types (n,e,n) represent different interaction for which we expect different models are needed
 - Example: (English, translate, French) requires a different model than (English, translate, Mandarin Chinese)
- The tradeoff: a heterogeneous graph is a more expressive representation, but it also has
 - increased computation and storage costs relative to homogeneous graphs
 - more complex implementation



Representing heterogeneous graphs in PyG

- HeteroData class supports creation of heterogeneous graph instances
- https://pytorch-geometric.readthedocs.io/en/latest/generated/torch_geometric.data.HeteroData.html#torch_geometric.data.HeteroData

HeteroData

```
class HeteroData ( _mapping: Optional[Dict[str, Any]] = None, **kwargs ) \[source\]
```

Bases: `BaseData`, `FeatureStore`, `GraphStore`

A data object describing a heterogeneous graph, holding multiple node and/or edge types in disjunct storage objects. Storage objects can hold either node-level, link-level or graph-level attributes. In general, `HeteroData` tries to mimic the behavior of a regular `nested` Python dictionary. In addition, it provides useful functionality for analyzing graph structures, and provides basic PyTorch tensor functionalities.

```
from torch_geometric.data import HeteroData

data = HeteroData()

# Create two node types "paper" and "author" holding a feature matrix:
data['paper'].x = torch.randn(num_papers, num_paper_features)
data['author'].x = torch.randn(num_authors, num_authors_features)

# Create an edge type "(author, writes, paper)" and building the
# graph connectivity:
data['author', 'writes', 'paper'].edge_index = ... # [2, num_edges]

data['paper'].num_nodes
>>> 23

data['author', 'writes', 'paper'].num_edges
>>> 52

# PyTorch tensor functionality:
data = data.pin_memory()
data = data.to('cuda:0', non_blocking=True)
```



Extending GCNs for heterogeneous graphs





Graph convolutional networks

- A GCN can be written as

$$\mathbf{h}_v^{(k+1)} = \sigma \left(\mathbf{W}^{(k)} \sum_{u \in N(v)} \frac{\mathbf{h}_u^{(k)}}{|N(v)|} \right)$$

NOTE: No self loop here, we'll address this later

- How to write this as aggregation $\rightarrow$ update?

$$\mathbf{m}_{N(v)}^{(k)} = \sum_{u \in N(v)} \frac{\mathbf{h}_u^{(k)}}{|N(v)|}$$

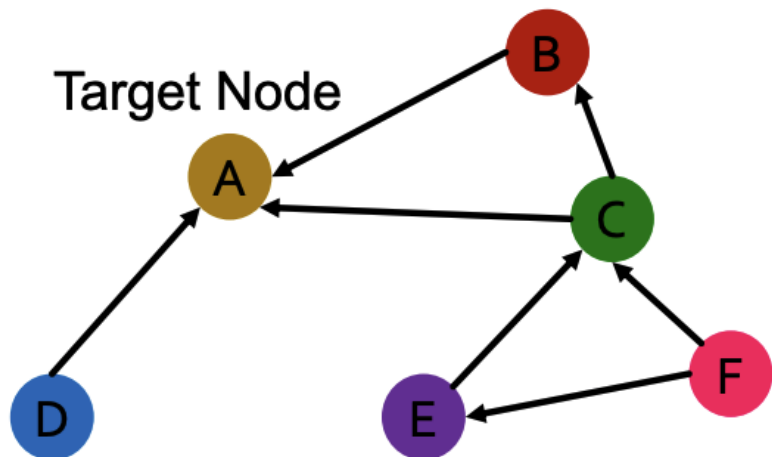
$$\mathbf{h}_v^{(k+1)} = \sigma \left(\mathbf{W}^{(k)} \mathbf{m}_{N(v)}^{(k)} \right)$$





Relational graph convolutional networks (RGCN)

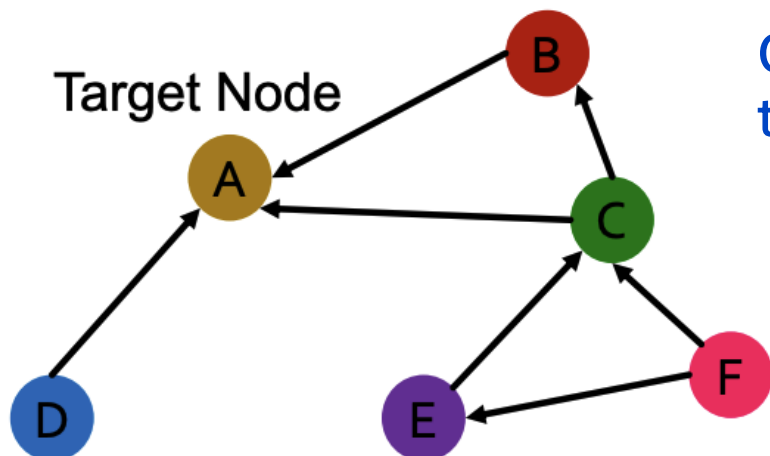
- Extend GCN to handle heterogeneous graphs with multiple relation types
- Start with a **directed** graph with **one** relation
 - How do we run GCN and update the representation of **target node A** on this graph?



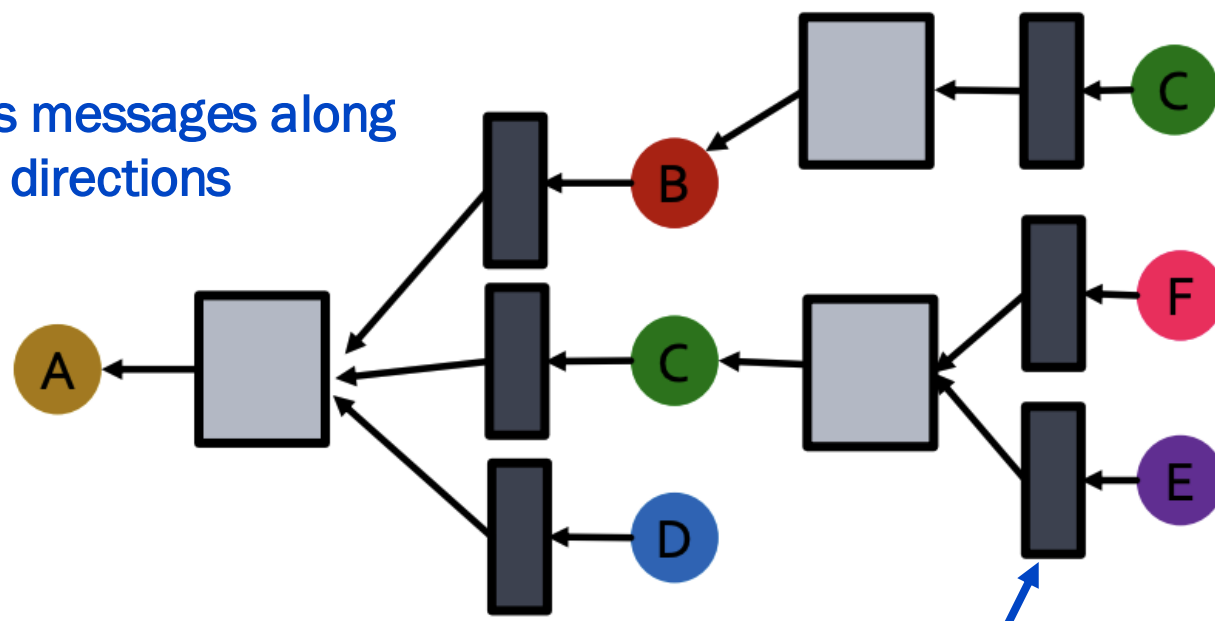


Relational graph convolutional networks (RGCN)

- Extend GCN to handle heterogeneous graphs with multiple relation types
- Start with a **directed** graph with **one** relation
 - How do we run GCN and update the representation of **target node A** on this graph?
 - This is implemented via the appropriate adjacency matrix



Only pass messages along
the edge directions

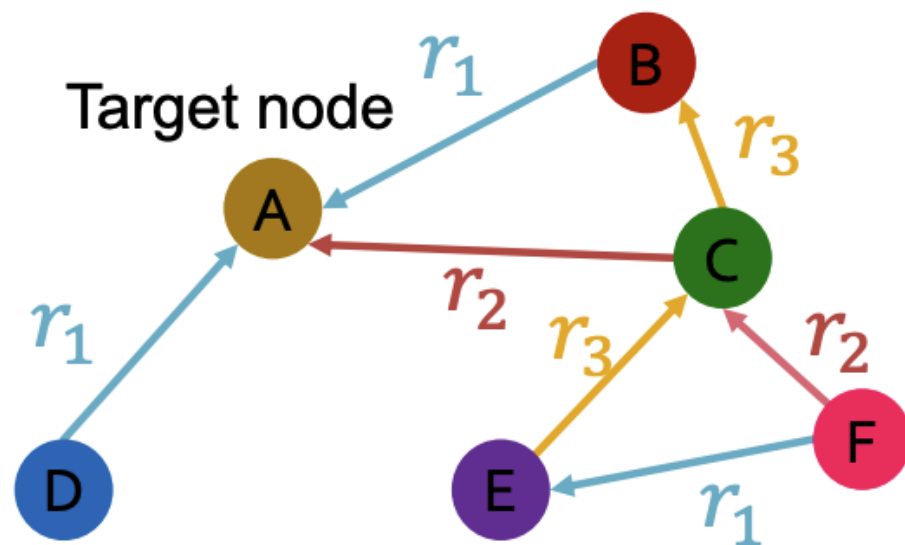


Same neural network across
all aggregations



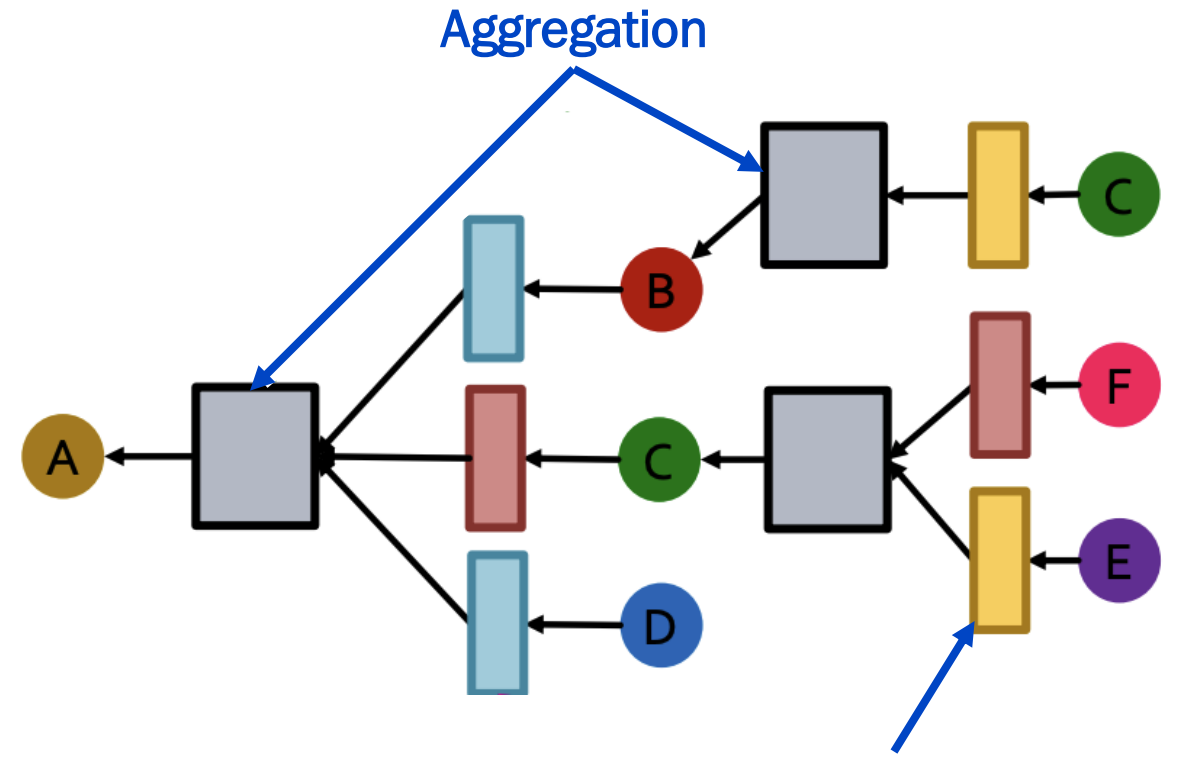
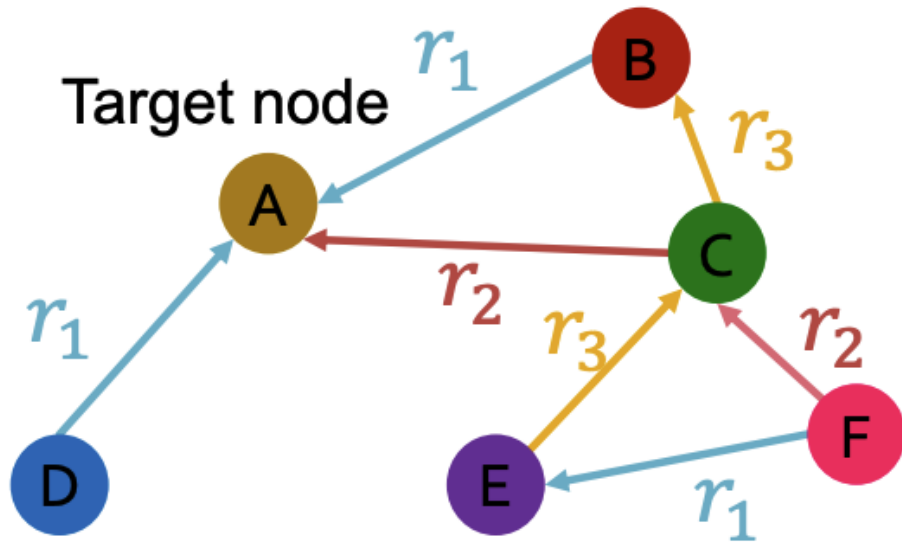
Relational graph convolutional networks (RGCN)

- What if the graph has multiple relation types?



Relational graph convolutional networks (RGCN)

- What if the graph has multiple relation types?
- Use different neural networks for different relation types



Distinct neural network for each relation type



Graph convolutional networks with self-loop matrix

- A GCN can be written as

$$\mathbf{h}_v^{(k+1)} = \sigma \left(\mathbf{W}^{(k)} \sum_{u \in N(v)} \frac{\mathbf{h}_u^{(k)}}{|N(v)|} \right)$$

- Let's add a self-loop with its own linear projection

$$\mathbf{h}_v^{(k+1)} = \sigma \left(\mathbf{W}^{(k)} \sum_{u \in N(v)} \frac{\mathbf{h}_u^{(k)}}{|N(v)|} + \mathbf{W}^{(k)} \mathbf{h}_v^{(k)} \right)$$



Relational graph convolutional networks (RGCN)

- RGCN includes a neural network for each relation type
- In the node update, it sums the node embeddings for each relation

$$\mathbf{h}_v^{(k+1)} = \sigma \left(\sum_{r \in R} \sum_{u \in N_r(v)} \frac{1}{c_{v,r}} \mathbf{w}_r^{(k)} \mathbf{h}_u^{(k)} + \mathbf{w}_0^{(k)} \mathbf{h}_v^{(k)} \right)$$

where $N_r(v)$ is the neighborhood of node v for relation type r and $c_{v,r} = |N_r(v)|$



RGCN PyG implementation

- The PyG [to_hetero](#) method converts a homogeneous GNN model into its heterogeneous equivalent

```
import torch
from torch_geometric.nn import SAGEConv, to_hetero

class GNN(torch.nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = SAGEConv((-1, -1), 32)
        self.conv2 = SAGEConv((32, 32), 32)

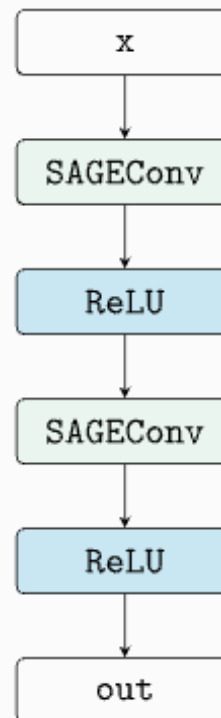
    def forward(self, x, edge_index):
        x = self.conv1(x, edge_index).relu()
        x = self.conv2(x, edge_index).relu()
        return x

model = GNN()

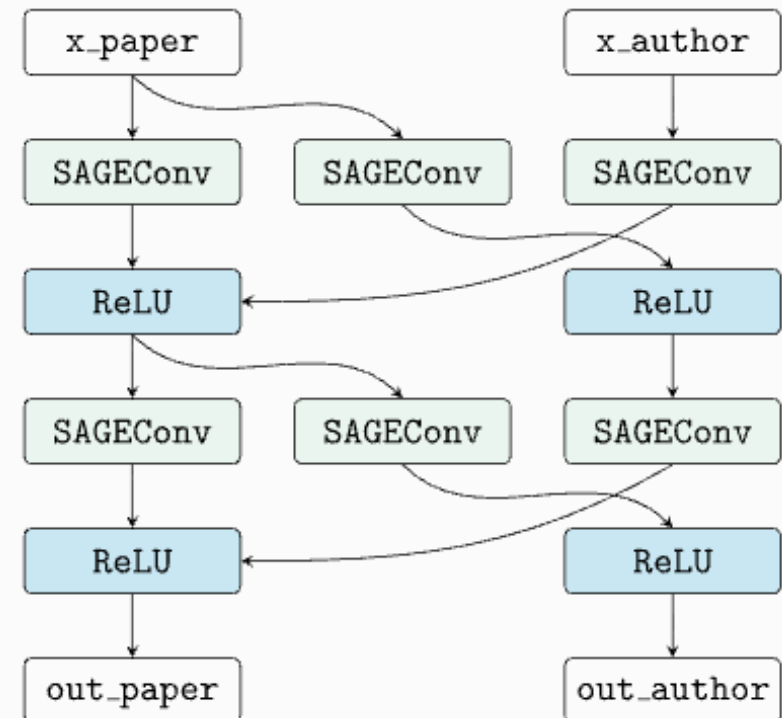
node_types = ['paper', 'author']
edge_types = [
    ('paper', 'cites', 'paper'),
    ('paper', 'written_by', 'author'),
    ('author', 'writes', 'paper'),
]
metadata = (node_types, edge_types)

model = to_hetero(model, metadata)
model(x_dict, edge_index_dict)
```

Homogeneous Model



Heterogeneous Model



x_dict and *edge_index_dict* are dictionaries that hold node feature and edge connection information for each node and edge type, respectively



RGCN parameter growth

- Each relation, type r , has L matrices (1 per GNN layer): $\mathbf{W}_r^{(1)}, \mathbf{W}_r^{(2)}, \dots, \mathbf{W}_r^{(L)}$
- The size of each $\mathbf{W}_r^{(k)}$ is $d^{k+1} \times d^k$
- **Rapid growth in the number of parameters w.r.t. the number of relations**
 - Model **complexity and overfitting** are concerns
- Consider two methods to regularize the weights, $\mathbf{W}_r^{(k)}$
 - Block diagonal matrices
 - Basis / dictionary learning

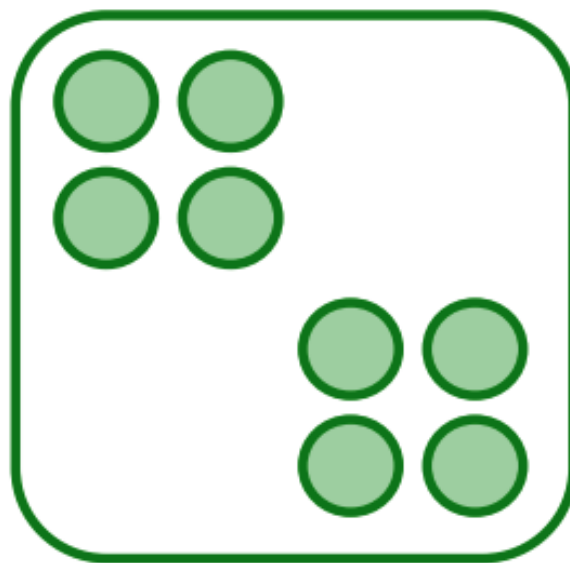


RGCN with block diagonal weight matrices

- Goals is to make the weight matrices sparse
- Make each $\mathbf{W}_r^{(k)}$ a **block diagonal matrix**
- If we use B blocks, the number of learning parameters reduces from $d^{(k+1)} \times d^{(k)}$ to $B \times \frac{d^{(k+1)}}{B} \times \frac{d^{(k)}}{B}$

$$\mathbf{W}_r^{(k)} =$$

Limitation: only nearby features can interact through $\mathbf{W}_r^{(k)}$





RGCN with block diagonal weight matrices

- Make each $\mathbf{W}_r^{(k)}$ a **block diagonal matrix**
- If we use B blocks, the number of learning parameters reduces from $d^{(k+1)} \times d^{(k)}$ to $B \times \frac{d^{(k+1)}}{B} \times \frac{d^{(k)}}{B}$
- Example
 - $\mathbf{W}_r$ with $d^{(k+1)} = d^{(k)} = 6$ and $B = 3$
 - Let's consider $\tilde{\mathbf{h}}_u = \mathbf{W}_r \mathbf{h}_u \in \mathbb{R}^{6 \times 1}$

$$\mathbf{W}_r^{(k)} = \begin{bmatrix} W_{11} & W_{12} & W_{13} & W_{14} & W_{15} & W_{16} \\ W_{21} & W_{22} & W_{23} & W_{24} & W_{25} & W_{26} \\ W_{31} & W_{32} & W_{33} & W_{34} & W_{35} & W_{36} \\ W_{41} & W_{42} & W_{43} & W_{44} & W_{45} & W_{46} \\ W_{51} & W_{52} & W_{53} & W_{54} & W_{55} & W_{56} \\ W_{61} & W_{62} & W_{63} & W_{64} & W_{65} & W_{66} \end{bmatrix} \quad \tilde{\mathbf{h}}_u = \begin{bmatrix} W_{11} \mathbf{h}_{u,1} + W_{12} \mathbf{h}_{u,2} + W_{13} \mathbf{h}_{u,3} + W_{14} \mathbf{h}_{u,4} + W_{15} \mathbf{h}_{u,5} + W_{16} \mathbf{h}_{u,6} \\ \vdots \\ W_{61} \mathbf{h}_{u,1} + W_{62} \mathbf{h}_{u,2} + W_{63} \mathbf{h}_{u,3} + W_{64} \mathbf{h}_{u,4} + W_{65} \mathbf{h}_{u,5} + W_{66} \mathbf{h}_{u,6} \end{bmatrix}$$

Every entry (feature) in $\tilde{\mathbf{h}}_u$ is influenced by **every** features in $\mathbf{h}_u$

Standard approach requires $d^{(k+1)} \times d^{(k)} = 36$ learning parameters



RGCN with block diagonal weight matrices

- Make each $\mathbf{W}_r^{(k)}$ a **block diagonal matrix**
- If we use B blocks, the number of learning parameters reduces from $d^{(k+1)} \times d^{(k)}$ to $B \times \frac{d^{(k+1)}}{B} \times \frac{d^{(k)}}{B}$
- Example
 - $\mathbf{W}_r$ with $d^{(k+1)} = d^{(k)} = 6$ and $B = 3$
 - Let's consider $\tilde{\mathbf{h}}_u = \mathbf{W}_r \mathbf{h}_u \in \mathbb{R}^{6 \times 1}$

Each feature is a transformation of a subset of the overall features (similar to a factor analysis)

$$\mathbf{W}_r^{(k)} = \begin{bmatrix} w_{11} & w_{12} & 0 & 0 & 0 & 0 \\ w_{21} & w_{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & w_{33} & w_{34} & 0 & 0 \\ 0 & 0 & w_{43} & w_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & w_{55} & w_{56} \\ 0 & 0 & 0 & 0 & w_{65} & w_{66} \end{bmatrix} \quad \tilde{\mathbf{h}}_u = \begin{bmatrix} w_{11} \mathbf{h}_{u,1} + w_{12} \mathbf{h}_{u,2} \\ w_{21} \mathbf{h}_{u,1} + w_{22} \mathbf{h}_{u,2} \\ w_{33} \mathbf{h}_{u,3} + w_{34} \mathbf{h}_{u,4} \\ w_{43} \mathbf{h}_{u,3} + w_{44} \mathbf{h}_{u,4} \\ w_{55} \mathbf{h}_{u,5} + w_{56} \mathbf{h}_{u,6} \\ w_{65} \mathbf{h}_{u,5} + w_{66} \mathbf{h}_{u,6} \end{bmatrix}$$

Standard approach requires $B \times \frac{d^{(k+1)}}{B} \times \frac{d^{(k)}}{B} = 12$ learning parameters

RGCN with block diagonal weight matrices

- Not currently implemented in PyG
- Implementation: do **not** create a sparse matrix
- Create the blocks and use batch matrix multiply
- In PyTorch / PyG the parameters would be created with `nn.Parameter` (see link below)

```
# Stack blocks into shape (B, block_size, block_size)
W_r_blocks = torch.stack([block_1, block_2, block_3])

# Reshape h_u from (6,) to (B, block_size)
h_u_resaped = h_u.view(3, 2) # Shape: (3, 2)

# Batched matrix-vector multiply
output = torch.bmm(W_r_blocks, h_u_resaped.unsqueeze(-1)).squeeze(-1)

# Flatten back to (6,)
output = output.view(-1)
```

<https://github.com/masino-teaching/ml4g/blob/main/notebooks/RGCN-with-block-diagonal-weights.ipynb>

```
In [3]: Wr = torch.stack([torch.rand(2,2) for _ in range(3)])

In [4]: Wr.shape
Out[4]: torch.Size([3, 2, 2])

In [5]: h = torch.rand(6,1)

In [6]: h
Out[6]:
tensor([[0.5272],
        [0.1507],
        [0.4108],
        [0.2431],
        [0.0395],
        [0.2747]])

In [7]: h_resaped = h.view(3,2)

In [8]: h_resaped
Out[8]:
tensor([[0.5272, 0.1507],
        [0.4108, 0.2431],
        [0.0395, 0.2747]])

In [9]: h_update = torch.bmm(Wr, h_resaped.unsqueeze(-1)).squeeze(-1)

In [10]: h_update
Out[10]:
tensor([[0.2952, 0.3379],
        [0.5286, 0.6133],
        [0.0317, 0.2690]])

In [11]: h_update = h_update.view(-1)

In [12]: h_update
Out[12]: tensor([0.2952, 0.3379, 0.5286, 0.6133, 0.0317, 0.2690])

In [13]: Wr
Out[13]:
tensor([[[[0.3617, 0.6938],
          [0.5346, 0.3718]],

         [[0.8256, 0.7796],
          [0.9789, 0.8691]],

         [[0.2915, 0.0735],
          [0.9806, 0.8382]]]])

In [14]: Wr[0].shape
Out[14]: torch.Size([2, 2])

In [15]: Wr[0,0,0]*h[0,0]+Wr[0,0,1]*h[1,0]
Out[15]: tensor(0.2952)
```



RCGN with basis learning

- Share weights across different relations
- Represent each relation matrix $\mathbf{W}_r^{(k)}$ as a linear combination of basis matrices

$$\mathbf{W}_r^{(k)} = \sum_{b=1}^B a_{r,b} \mathbf{V}_b^{(k)}$$

where $\mathbf{V}_b^{(k)}$ are basis matrices and $a_{r,b}$ is a scalar weighting

- Now each relation only adds B learning parameters $\{a_{r,1}, \dots, a_{r,B}\}$

RGCN PyG implementation

- The PyG [to_hetero_with_bases](#) method converts a homogeneous GNN model into its heterogeneous equivalent

```
import torch
from torch_geometric.nn import SAGEConv, to_hetero_with_bases

class GNN(torch.nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = SAGEConv((16, 16), 32)
        self.conv2 = SAGEConv((32, 32), 32)

    def forward(self, x, edge_index):
        x = self.conv1(x, edge_index).relu()
        x = self.conv2(x, edge_index).relu()
        return x

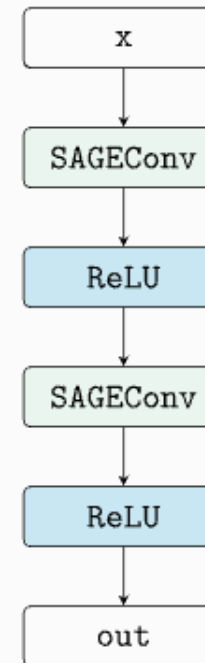
model = GNN()

node_types = ['paper', 'author']
edge_types = [
    ('paper', 'cites', 'paper'),
    ('paper', 'written_by', 'author'),
    ('author', 'writes', 'paper'),
]
metadata = (node_types, edge_types)

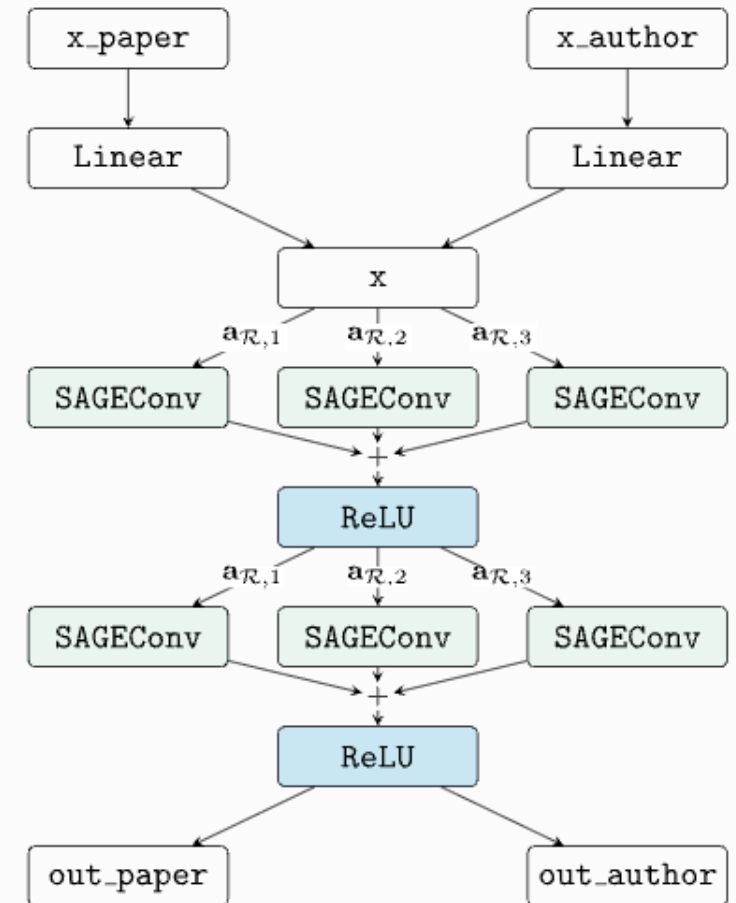
model = to_hetero_with_bases(model, metadata, num_bases=3,
                             in_channels={'x': 16})
model(x_dict, edge_index_dict)
```

x_dict and $edge_index_dict$ are dictionaries that hold node feature and edge connection information for each node and edge type, respectively

Homogeneous Model



Heterogeneous Model





Inferences on relational graphs





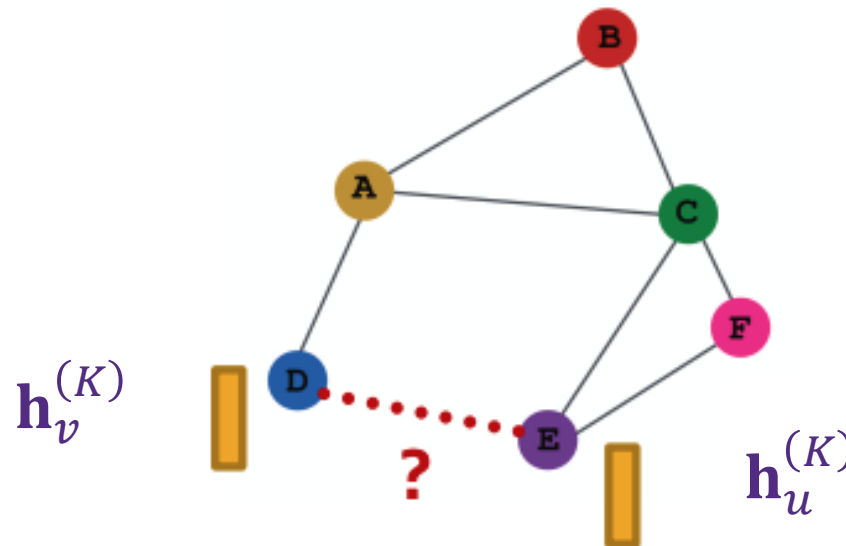
Heterogeneous graph / node classification

- Goal: Infer the label of a given node
- Use the node embedding, $\mathbf{h}_v^{(K)} \in \mathbb{R}^{d_K}$ from the final RGCN layer, K
- Pass this embedding through the prediction head
 - Example: linear layer $\mathbf{W}_{\tau(v)}^{out} \in \mathbb{R}^{(d_K \times C)}$ where C is the number of classes and $\mathbf{h}_v^{(K)} \mathbf{W}_{\tau(v)}^{out}$ are the logits
 - Note: there is a separate matrix for each node type $\tau(v)$
- For heterogeneous graph classification:
 - Create a prediction head for each node type, $\tau(v)$, to predict class label (e.g., global mean pooling $\rightarrow$ linear layer)
 - Aggregate (e.g., mean) predictions across prediction heads



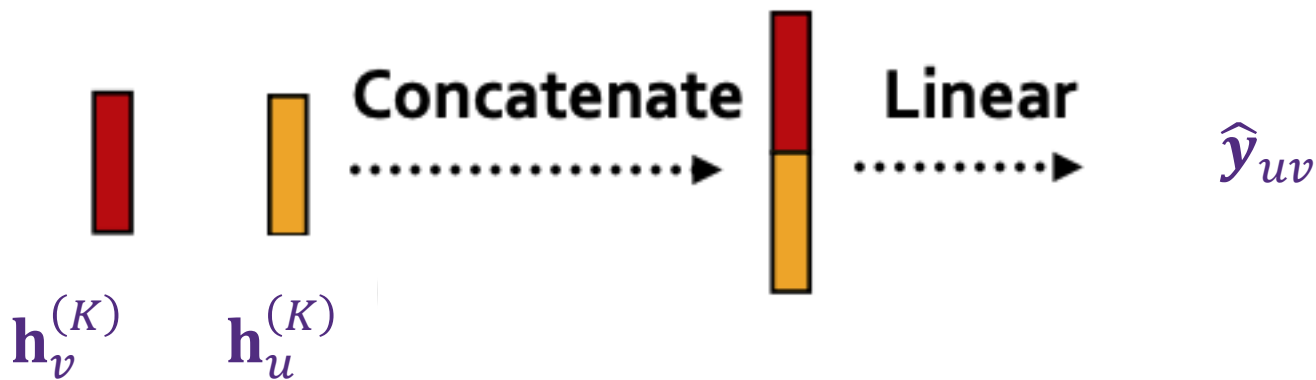
Edge-level prediction head

- Use **pairs** of node embeddings as input to a prediction function
- $\hat{y}_{uv} = g(\mathbf{h}_v^{(K)}, \mathbf{h}_u^{(K)})$
- What are the options for $g(\mathbf{h}_v^{(K)}, \mathbf{h}_u^{(K)})$?



Edge-level prediction head with concatenation

- One option is to concatenate the u and v node embeddings
- Prediction function, g , operates on the concatenated vector
- Example: g is a linear layer, $\hat{y}_{uv} = g \left(\left[\mathbf{h}_u^{(K)} \mid \mathbf{h}_v^{(K)} \right] \right) = W^g \left[\mathbf{h}_u^{(K)} \mid \mathbf{h}_v^{(K)} \right]$



- Here, $W^g \in \mathbb{R}^{C \times 2d_K}$ maps node embeddings from $\mathbb{R}^{2d_K}$ to $\mathbb{R}^C$

Account for the concatenation
of embeddings



Edge-level prediction head with dot product

- The prediction head function is just the dot product of the node embeddings
 - $\hat{y}_{uv} = \mathbf{h}_u^{(K)T} \cdot \mathbf{h}_v^{(K)}$
 - This **only applies to 1-way prediction** (i.e., binary classification, single regression, link prediction)
- Applying to **C-way** prediction with a **learnable generalized dot product**
 - Learn c weight matrices

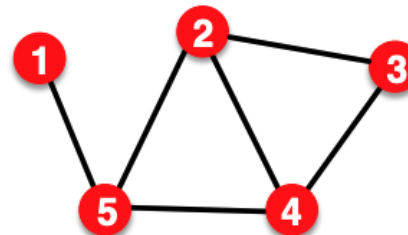
$$\begin{aligned}\hat{y}_{uv,1} &= \mathbf{h}_u^{(K)T} \mathbf{W}^1 \mathbf{h}_v^{(K)} \\ &\vdots \\ \hat{y}_{uv,c} &= \mathbf{h}_u^{(K)T} \mathbf{W}^c \mathbf{h}_v^{(K)} \\ \hat{\mathbf{y}}_{uv} &= [\hat{y}_{uv,1}, \dots, \hat{y}_{uv,c}] \in \mathbb{R}^c\end{aligned}$$

where $\mathbf{W}^c \in \mathbb{R}^{d_K \times d_K}$

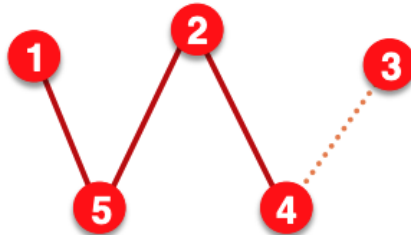


Setting up edge prediction

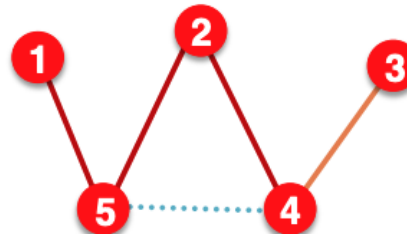
- Transductive link prediction split



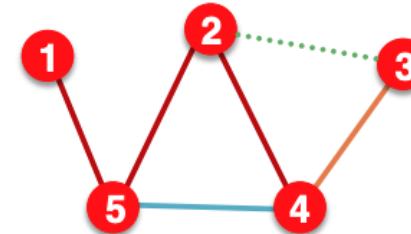
The original graph



(1) At training time:
Use **training message edges** to predict **training supervision edges**



(2) At validation time:
Use **training message edges & training supervision edges** to predict **validation edges**

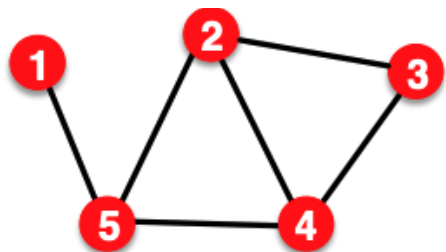


(3) At test time:
Use **training message edges & training supervision edges & validation edges** to predict **test edges**



Setting up edge prediction

- Edge prediction split



The original graph

Split
→



Split Graph with 4 categories of edges

Every edge has a relation type, independent of the four (4) categories of splits

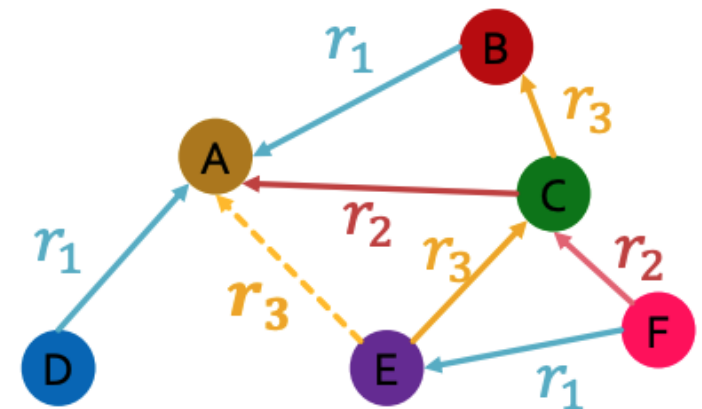
For a heterogeneous graph, the homogeneous graphs formed by a given relation, has four (4) splits





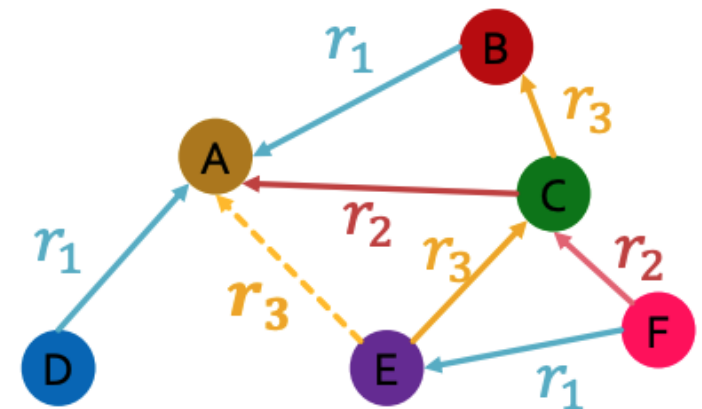
RGCN for edge prediction

- Assume (E, r_3, A) is a **training supervision edge**, and all other edges are **training message edges**
 - Other edges (not shown yet) are validation or test supervision edges
- Use the RGCN to predict probability for edge (E, r_3, A)
 - Take the final RGCN layer embeddings, $\mathbf{h}_E^{(K)}$ and $\mathbf{h}_A^{(K)}$, for E and A
 - Apply a relation specific prediction head $g_r: \mathbb{R}^{d_K \times d_K} \rightarrow \mathbb{R}$
 - Example: $g_{r_3} = \left(\mathbf{h}_E^{(K)}\right)^T \mathbf{W}_{r_3} \mathbf{h}_A^{(K)}$ where $\mathbf{W}_{r_3} \in \mathbb{R}^{d_K \times d_K}$



RGCN training for edge prediction

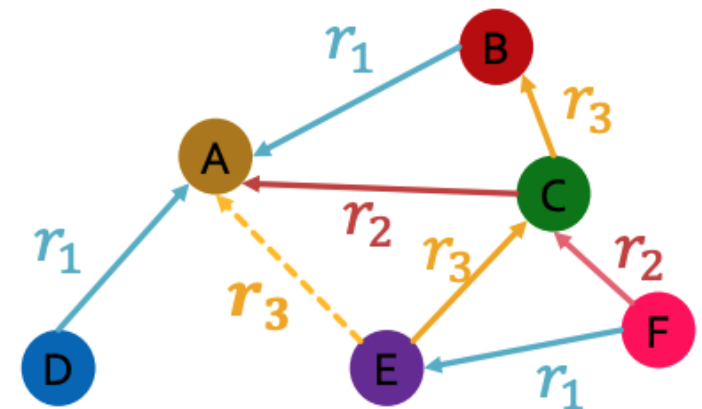
1. Use the RGCN to score the **training supervision edge** (E, r_3, A)
 - Message passing occurs only on the training message edges for this relation type
2. Create negative edge(s) by perturbing the **supervision edge** (E, r_3, A) , for example by changing the tail: (E, r_3, B) , (E, r_3, F)
 - Negative edges should not be in the actual graph (i.e., not a member of any of the 4 splits)



RGCN training for edge prediction

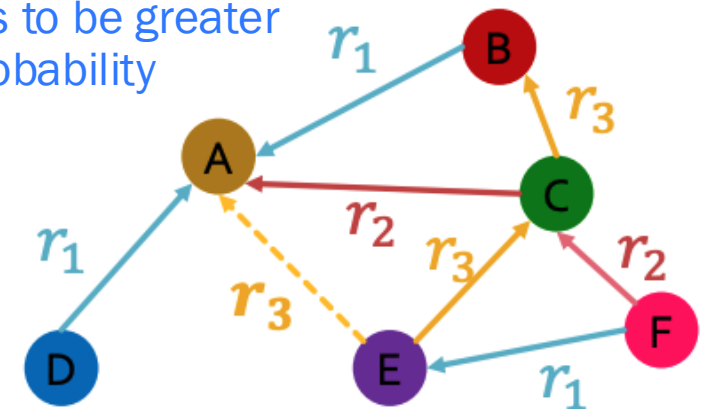
1. Use the RGCN to score the **training supervision edge** (E, r_3, A)
 - Message passing occurs only on the training message edges for this relation type
2. Create **negative edge(s)** by perturbing the **supervision edge** (E, r_3, A) , for example by changing the tail: (E, r_3, B) , (E, r_3, F)
 - Negative edges should not be in the actual graph (i.e., not a member of any of the 4 splits)
3. Score the **negative edge** with the RGCN
4. Optimize cross-entropy loss function

$$l = -\log \sigma \left(g_{r_3} \left(\mathbf{h}_E^{(K)}, \mathbf{h}_A^{(K)} \right) \right) - \log \left(1 - \sigma \left(g_{r_3} \left(\mathbf{h}_E^{(K)}, \mathbf{h}_B^{(K)} \right) \right) \right)$$



RGCN evaluation for edge prediction

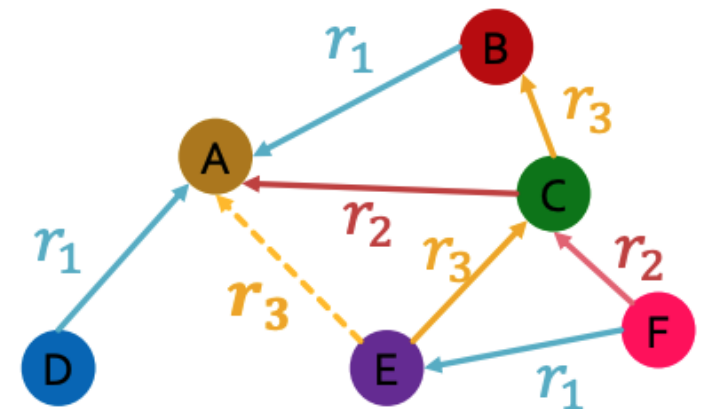
- How should we evaluate model performance?
- We could use standard **binary classification performance metrics**, but these **do not align with the typical use case**
 - Most applications involve very large, sparse graphs
 - Nearly all “edge” predictions are negative
 - The typical requirement is “precision”, more specifically, if the model indicates there are K predicted edges that are “missing” from the graph, how many of these are valid?
- **Intuition: we want the score for validation / test supervision edges to be greater than negative samples, without worrying about the normalized probability**





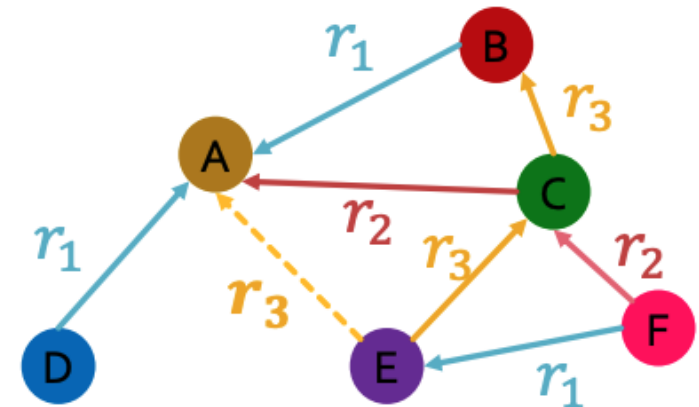
RGCN evaluation for edge prediction

- Consider evaluation during validation (same procedure for test)
- Assume (E, r_3, D) is a **validation supervision edge**
- Use the RGCN to score (E, r_3, D) and sampled negative edges, let's assume those are (E, r_3, B) , (E, r_3, F)
- Rank the edges by their RGCN score
- Obtain the rank, R_K , of the (E, r_3, D) , where K is the number of edges considered
- Repeat this for all validation edges
 - Compute *Hits@k* – that is, how many of the validation supervision edges have a rank of $k < K$ or better? (Higher number of “hits” is better)
 - Reciprocal rank $\frac{1}{R_K}$ (Higher is better – indicates lower R_K)



RGCN evaluation for edge prediction

- Consider evaluation during validation (same procedure for test)
- Assume (E, r_3, D) is a **validation supervision edge**
- Use the RGCN to score (E, r_3, D) and sampled negative edges, let's assume those are (E, r_3, B) , (E, r_3, F)
 - Here $K = 3$ Let's consider Hits@2 (i.e, $k = 2$)
- Assume we get following ordering
 - $g_{r_3}(E, r_3, D)$ R_K of is 1 (E, r_3, D)
 - Increment hits@2 by 1
 - Reciprocal rank is 1 for this sample
 - $g_{r_3}(E, r_3, B)$
 - $g_{r_3}(E, r_3, F)$



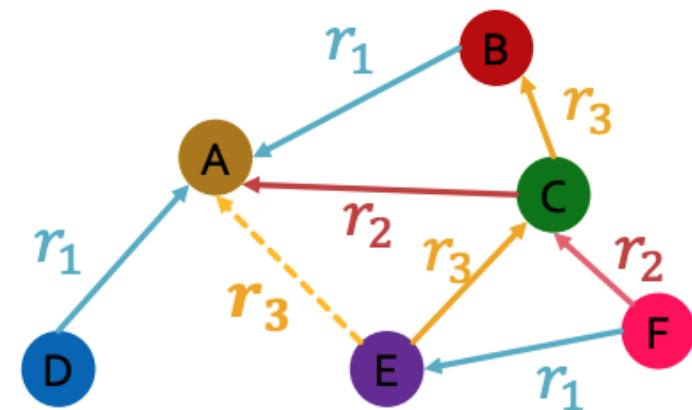


RGCN evaluation for edge prediction

- Consider evaluation during validation (same procedure for test)
- Assume (E, r_3, D) is a **validation supervision edge**
- Use the RGCN to score (E, r_3, D) and sampled negative edges, let's assume those are (E, r_3, B) , (E, r_3, F)
 - Here $K = 3$ Let's consider Hits@2 (i.e, $k = 2$)
- Assume we get following ordering
 - $g_{r_3}(E, r_3, B)$
 - $g_{r_3}(E, r_3, D)$
 - $g_{r_3}(E, r_3, F)$

R_K of is 2 (E, r_3, D)

- Increment hits@2 by 1
- Reciprocal rank is 1/2 for this sample

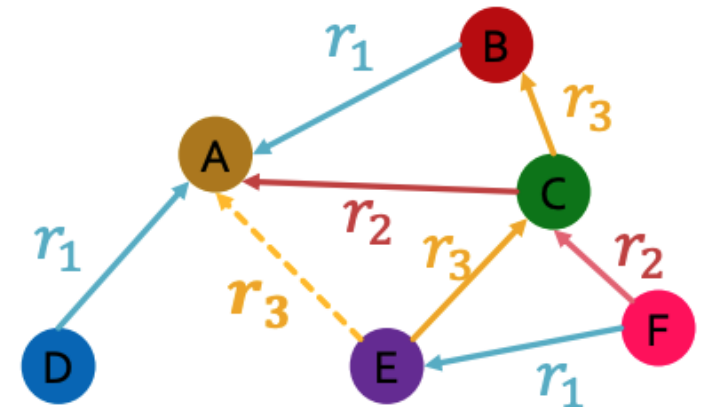


RGCN evaluation for edge prediction

- Consider evaluation during validation (same procedure for test)
- Assume (E, r_3, D) is a **validation supervision edge**
- Use the RGCN to score (E, r_3, D) and sampled negative edges, let's assume those are (E, r_3, B) , (E, r_3, F)
 - Here $K = 3$ Let's consider Hits@2 (i.e, $k = 2$)
- Assume we get following ordering
 - $g_{r_3}(E, r_3, B)$
 - $g_{r_3}(E, r_3, F)$
 - $g_{r_3}(E, r_3, D)$

R_K of is 3 (E, r_3, D)

- Increment hits@2 by 0 (**no credit for this sample**)
- Reciprocal rank is 1/3 for this sample



Benchmarking heterogeneous graph methods

Numerous heterogeneous graph datasets in PyG

Heterogeneous Datasets

DBP15K	The DBP15K dataset from the " Cross-lingual Entity Alignment via Joint Attribute-Preserving Embedding " paper, where Chinese, Japanese and French versions of DBpedia were linked to its English version.
AMiner	The heterogeneous AMiner dataset from the " metapath2vec: Scalable Representation Learning for Heterogeneous Networks " paper, consisting of nodes from type "paper", "author" and "venue".
OGB_MAG	The <code>ogbn-mag</code> dataset from the " Open Graph Benchmark: Datasets for Machine Learning on Graphs " paper.
DBLP	A subset of the DBLP computer science bibliography website, as collected in the " MAGNN: Metapath Aggregated Graph Neural Network for Heterogeneous Graph Embedding " paper.
MovieLens	A heterogeneous rating dataset, assembled by GroupLens Research from the MovieLens web site , consisting of nodes of type "movie" and "user".
MovieLens100K	The MovieLens 100K heterogeneous rating dataset, assembled by GroupLens Research from the MovieLens web site , consisting of movies (1,682 nodes) and users (943 nodes) with 100K ratings between them.
MovieLens1M	The MovieLens 1M heterogeneous rating dataset, assembled by GroupLens Research from the MovieLens web site , consisting of movies (3,883 nodes) and users (6,040 nodes) with approximately 1 million ratings between them.
IMDB	A subset of the Internet Movie Database (IMDB), as collected in the " MAGNN: Metapath Aggregated Graph Neural Network for Heterogeneous Graph Embedding " paper.
LastFM	A subset of the last.fm music website keeping track of users' listening information from various sources, as collected in the " MAGNN: Metapath Aggregated Graph Neural "





Summary of RGCN

- RGCN extends GCN to heterogeneous graphs
- Can perform node / graph classification and link prediction
- Ideas can be extended to other RGNN models such as RGraphSAGE, RGIN, RGAT

